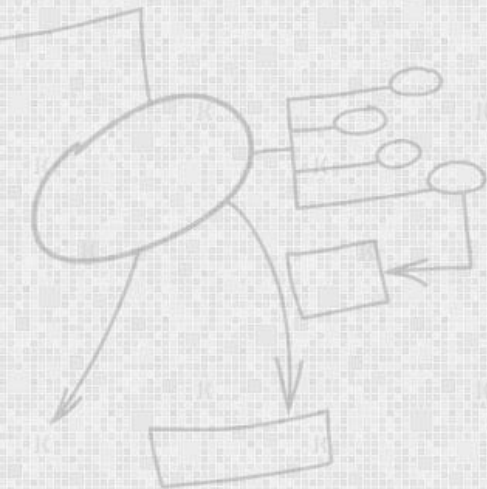


EC 240 Database Engineering

Lecture 1: Introduction

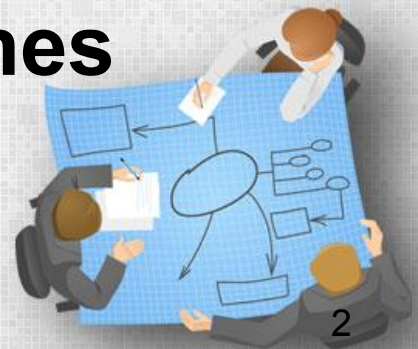
By

Dr Wasi Haider Butt



Today's Lecture: Agenda

- **Introduction**
- **Course outlines**
- **Basic concepts**
- **Comparison of file processing systems with the database systems**
- **Database development approaches**



Course Outline

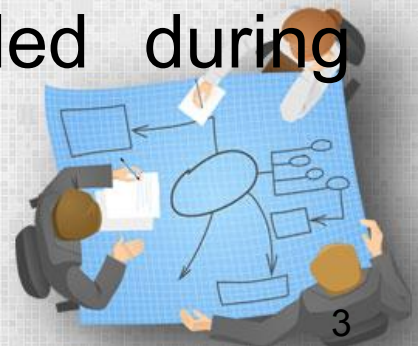
- **Text Book:**

Modern Database Management, by Hoffer,
Prescott, MsFadden, Latest Edition

References:

Fundamentals of Database Systems by Elmasri,
Navathi

Some other material may be added during
course



Course Outline

- Theoretical Background
- Database design and development
- Conceptual design
- Relational design
- Physical design
- Relational algebra
- SQL for database querying
- Normalization
- Web/Windows/Mobile based applications development



Course Outline

- Query optimization
- Distributed databases
- No SQL Databases
- Transactions
- Introduction to data warehouse and data mining/Big Data
- Online analytical processing (OLAP)



Marks Distribution (Tentative)

• Quizzes	10	
• Project	10	
• Sessional Exams		30
• Assignments	10	
• Final	40	



Some basic concepts

- Data
 - Structured
 - Un-Structured
- Information
- Metadata
- Information Systems
 - Data Part
 - App Part



Database (Definition)

In the broadest sense, a database is anything that stores data. A phone book, for instance, could be considered a database as it stores related pieces of information such as name and phone number.

However, in the world of computers, a database usually refers to a **collection of related pieces of information** stored electronically. Aside from the ability to **store data**, a database also provides a way for other computer programs to **quickly retrieve and update** desired pieces of data.

Reference:

<http://www.mariosalexandrou.com/definition/database.asp>

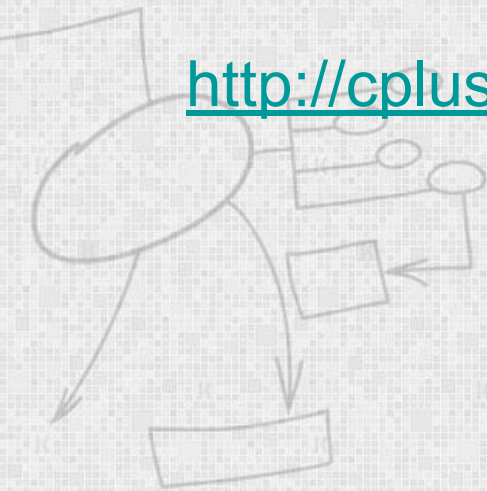


Database (Definition)

A database is an application that manages data and allows fast storage and retrieval of that data.

Reference

<http://cplus.about.com/od/glossar1/g/databasedefn.htm>



Database (Definition)

A database is a collection of information that is organized so that it can easily be accessed, managed, and updated.

Reference:

http://searchsqlserver.techtarget.com/sDefinition/0,,sid87_gci211895,00.html



Definitions (By Hoffer)

- Database (Concluded): **organized collection** of **logically related** data

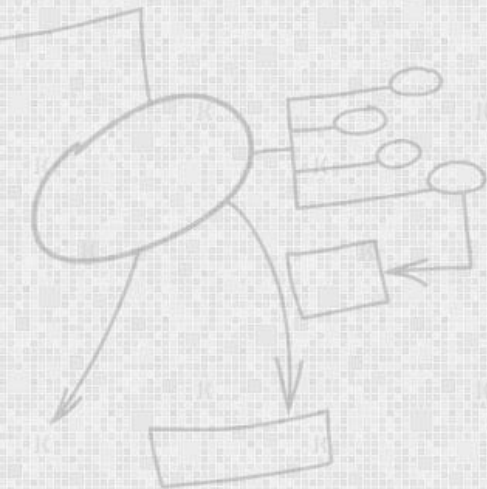
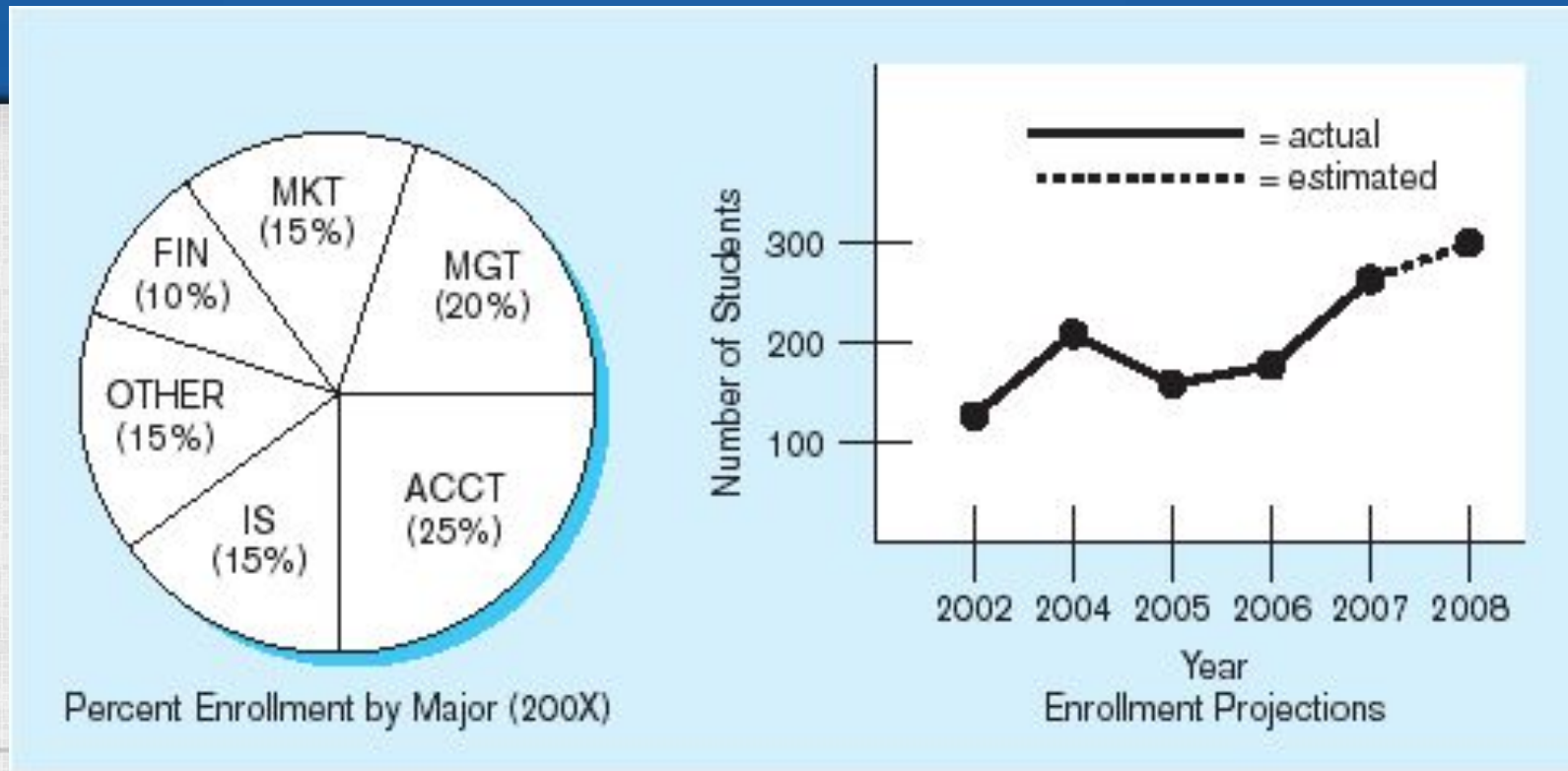


Figure 1-1b Summarized data



Graphical displays turn data into useful information that managers can use for decision making and interpretation



Table 1-1 Example Metadata for Class Roster

<i>Data Item</i>			<i>Value</i>			
Name	Type	Length	Min	Max	Description	Source
Course	Alphanumeric	30			Course ID and name	Academic Unit
Section	Integer	1	1	9	Section number	Registrar
Semester	Alphanumeric	10			Semester and year	Registrar
Name	Alphanumeric	30			Student name	Student IS
ID	Integer	9			Student ID (SSN)	Student IS
Major	Alphanumeric	4			Student major	Student IS
GPA	Decimal	3	0.0	4.0	Student grade point average	Academic Unit

Descriptions of the properties or characteristics of the data, including data types, field sizes, allowable values, and data context



Disadvantages of File Processing

- **Program-Data Dependence**
 - All programs maintain metadata for each file they use
- **Duplication of Data**
 - Different systems/programs have separate copies of the same data
- **Limited Data Sharing**
 - No centralized control of data
- **Lengthy Development Times**
 - Programmers must design their own file formats
- **Excessive Program Maintenance**
 - 80% of information systems budget



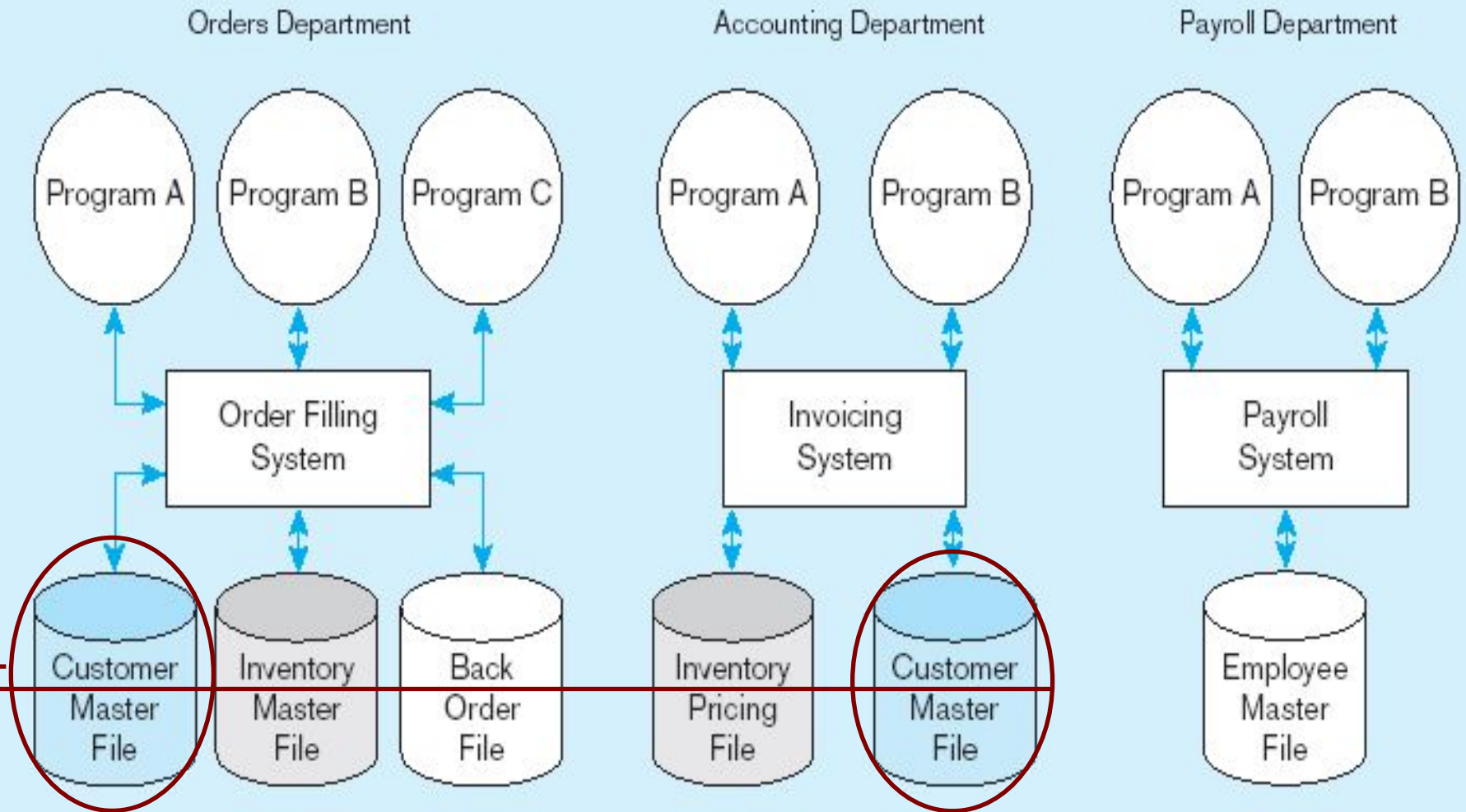
Problems with Data Dependency

- Each application programmer **must maintain** his/her **own data**
- Each application program needs to **include code** for the **metadata** of each file
- Each application program must have its **own processing routines** for reading, inserting, updating, and deleting data
- **Lack of coordination and central control**
- **Non-standard file formats**



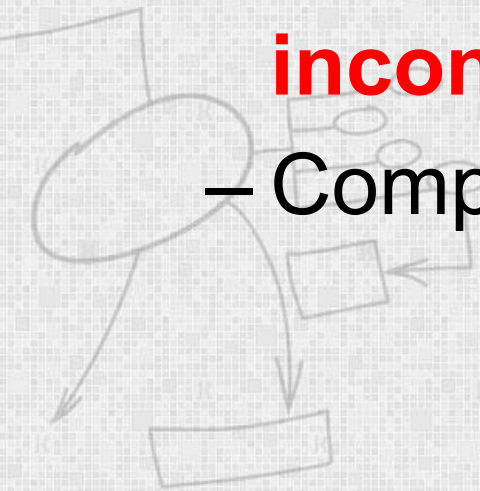
Figure 1-3 Old file processing systems at Pine Valley Furniture Company

Duplicate Data



Problems with Data Redundancy

- **Waste** of **space** to have duplicate data
- Causes more **maintenance** headaches
- The biggest problem:
 - Data changes in one file could cause **inconsistencies**
 - Compromises in ***data integrity***

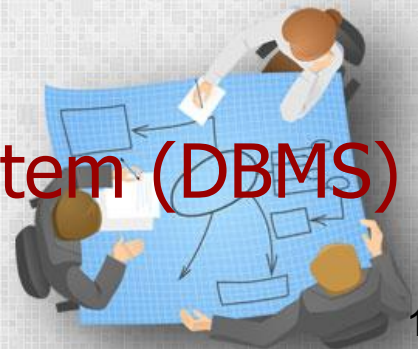


SOLUTION:

The DATABASE Approach

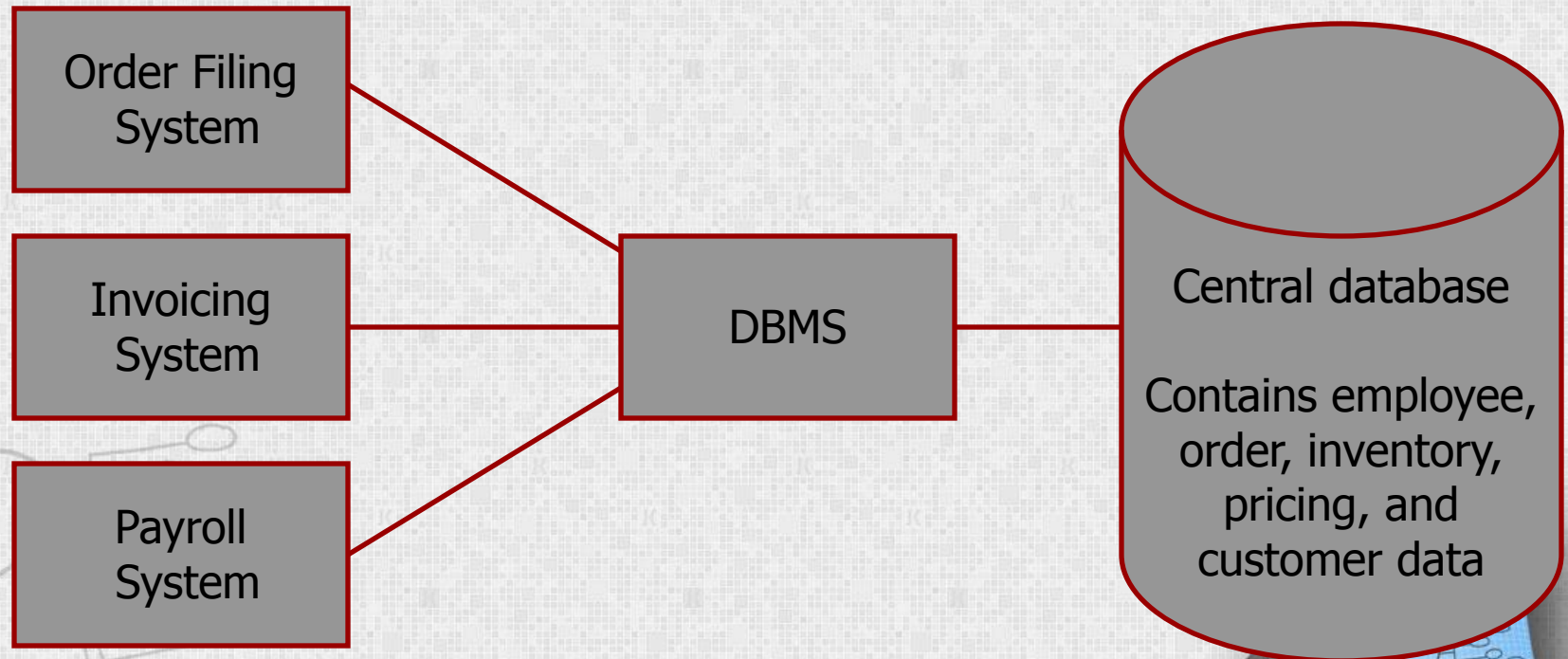
- Central repository of shared data
- Data is managed by a controlling agent
- Stored in a standardized, convenient form

Requires a Database Management System (DBMS)



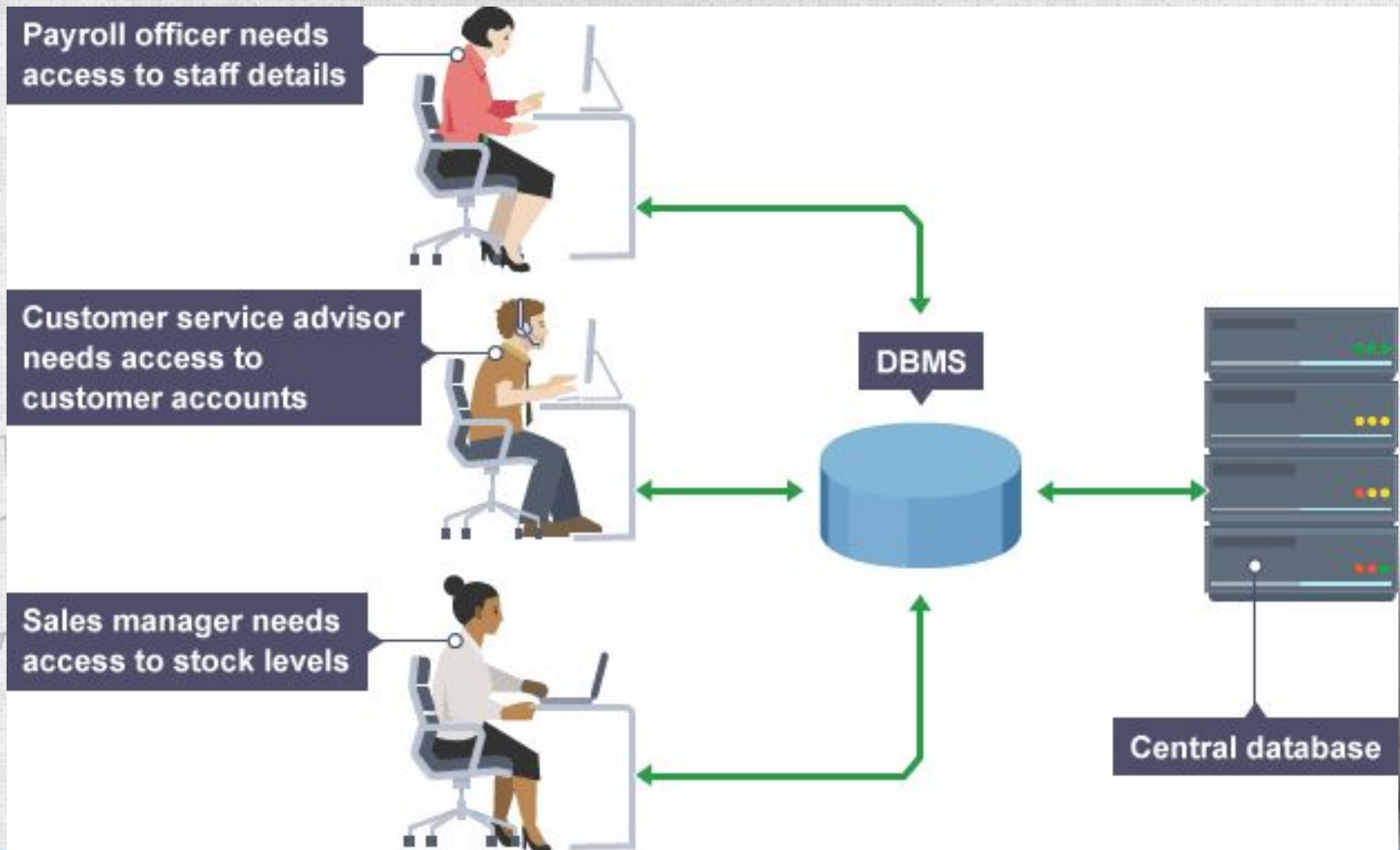
Database Management System

- A software system that is used to create, maintain, and provide controlled access to user databases



DBMS manages data resources like an operating system manages hardware resources

Database Management System



Database Management System



Advantages of the Database Approach

- Program-data **independence**
- **Planned** data redundancy
- Improved data **consistency**
- Improved data **sharing**
- Increased application development **productivity**
- Enforcement of **standards**
- Improved data **quality**
- Improved data **accessibility** and **responsiveness**
- Reduced program **maintenance**
- Improved **decision** support

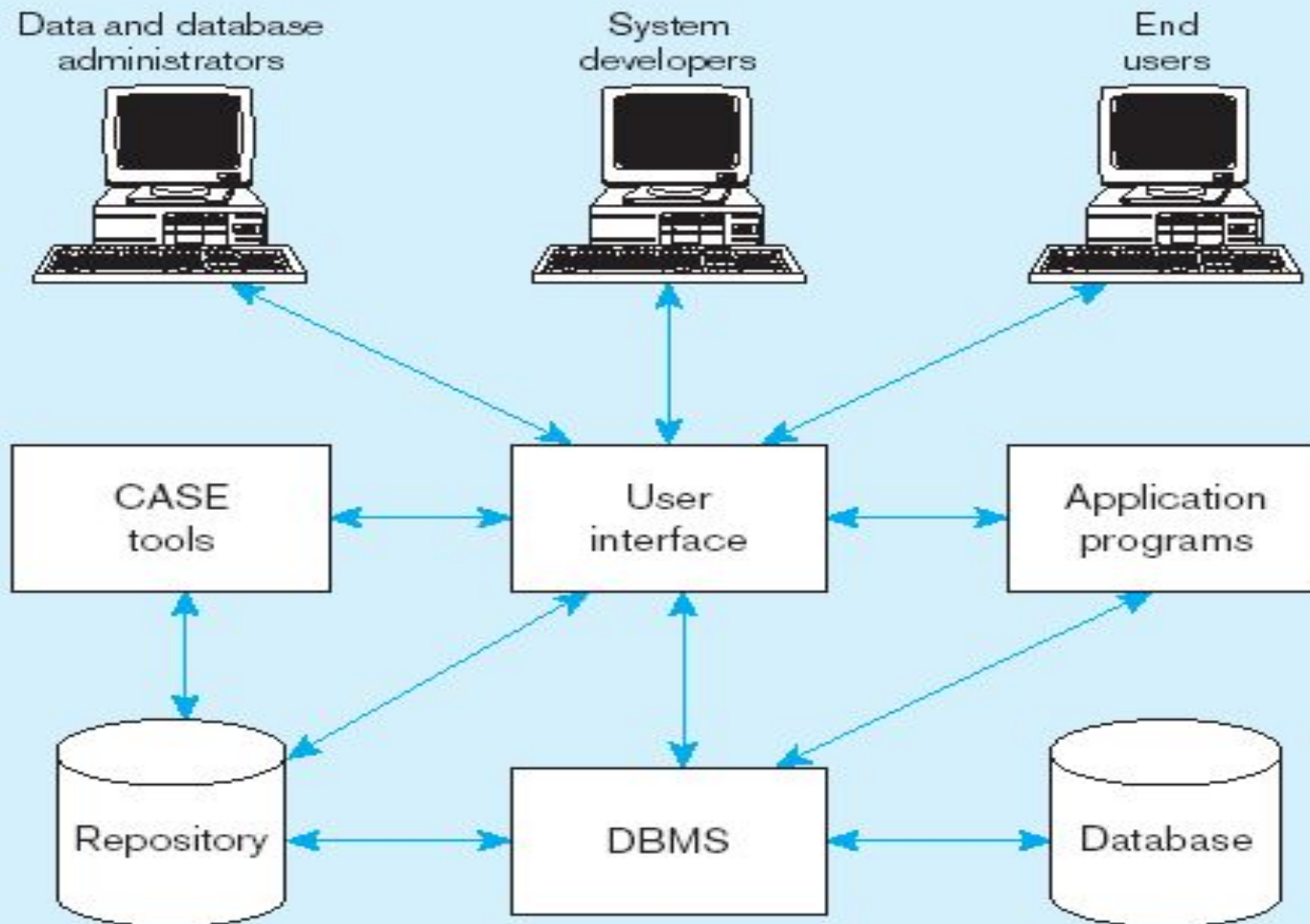


Costs and Risks of the Database Approach

- New, specialized personnel
- Installation and management cost and complexity
- Conversion costs
- Need for explicit backup and recovery
- Organizational conflict



Figure 1-5 Components of the Database Environment



Components of the Database Environment

- **CASE Tools**—computer-aided software engineering
- **Repository**—centralized storehouse of metadata
- **Database Management System (DBMS)** —software for managing the database
- **Database**—storehouse of the data
- **Application Programs**—software using the data
- **User Interface**—text and graphical displays to users
- **Data/Database Administrators**—personnel responsible for maintaining the database
- **System Developers**—personnel responsible for designing databases and software
- **End Users**—people who use the applications and databases



The Range of Database Applications

- Personal databases
- Workgroup databases
- Departmental/divisional databases
- Enterprise database

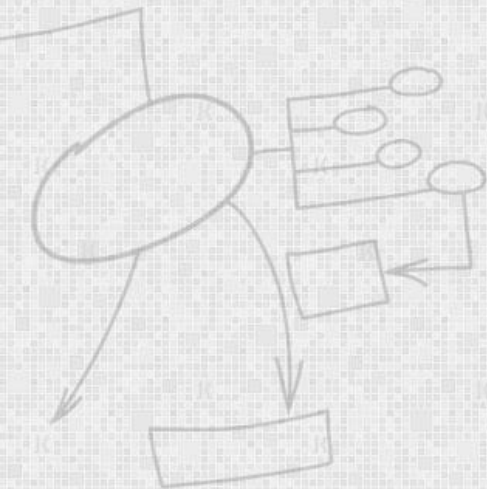
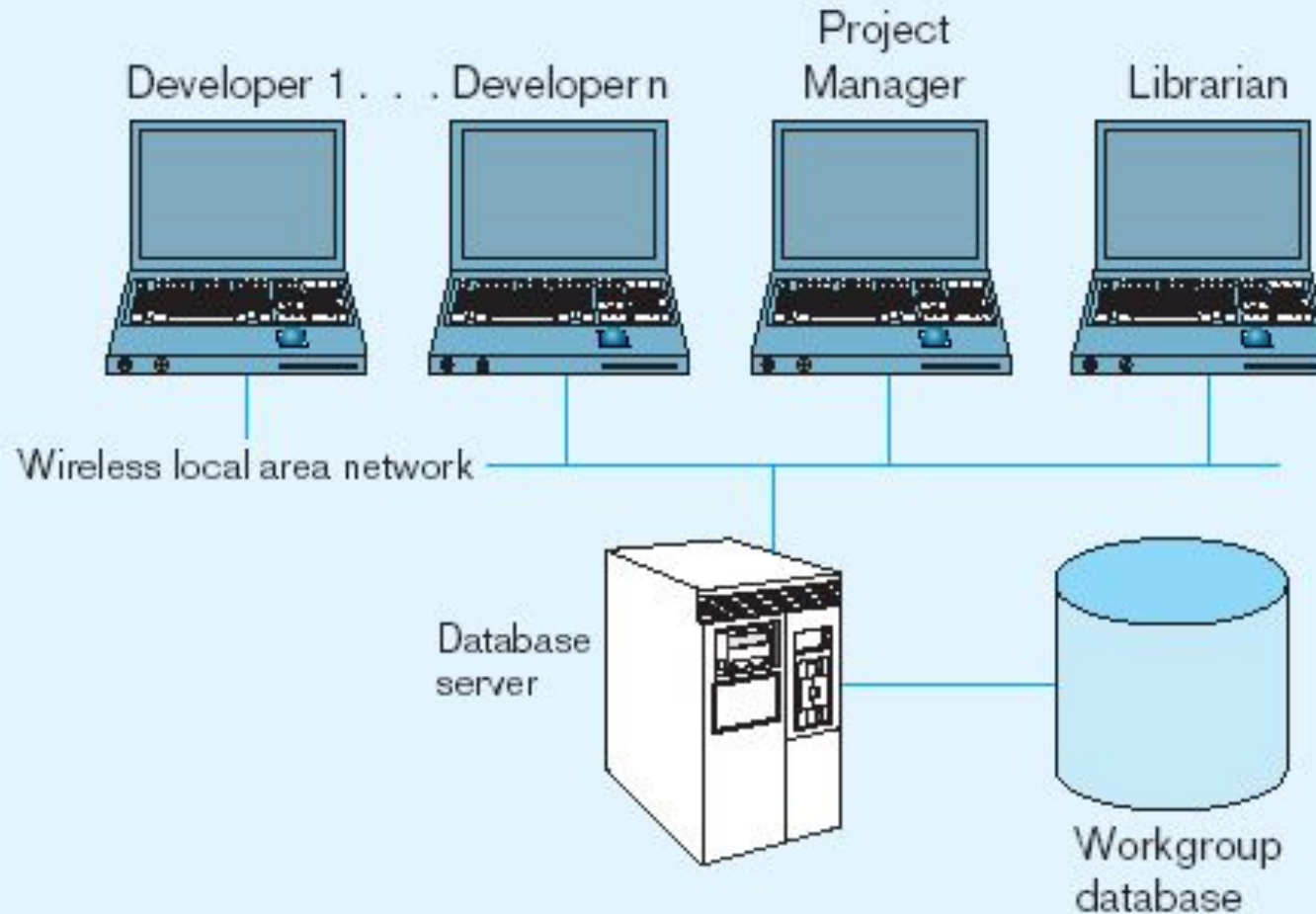


Table 1-6 Summary of Database Applications (adapted from White, 1995)

<i>Type of Database</i>	<i>Typical Number of Users</i>	<i>Typical Architecture</i>	<i>Typical Size of Database</i>
Personal	1	Desktop/laptop computer, PDA	Megabytes
Workgroup	5–25	Client/server (two-tier)	Megabytes–gigabytes
Department/Division	25–100	Client/server (three-tier)	Gigabytes
Enterprise	>100	Client/server (distributed or parallel server)	Gigabytes–terabytes
Web-enabled	>1000	Web server and application servers	Megabytes–gigabytes



Figure 1-7 Workgroup database with wireless local area network

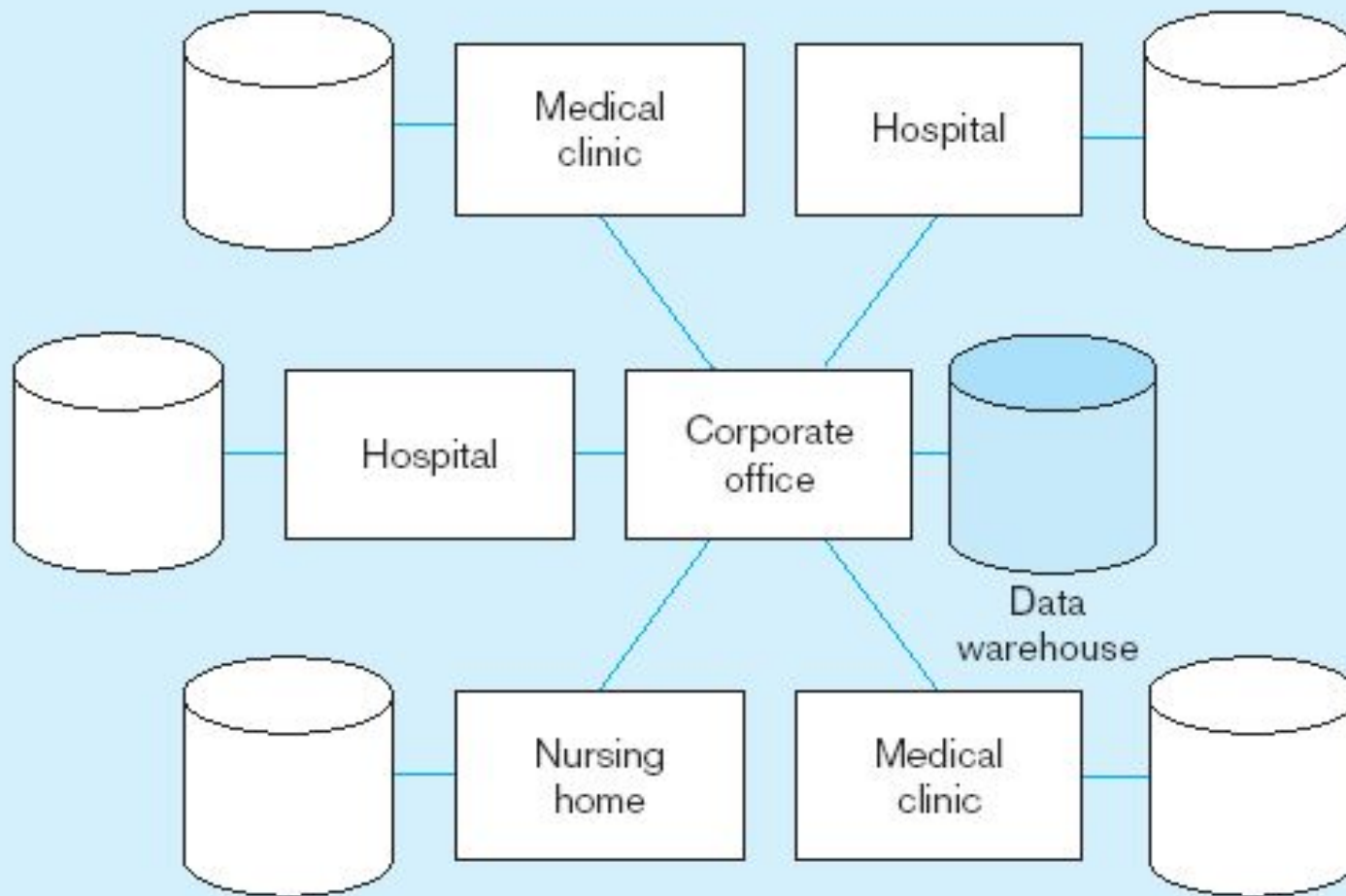


Enterprise Database Applications

- Enterprise Resource Planning (ERP)
 - Integrate all enterprise business functions (manufacturing, finance, sales, marketing, inventory, accounting, human resources)
- Data Warehouse
 - Integrated decision support system derived from various operational databases
 - Operational VS Non-Operational Data stores



Figure 1-8 An enterprise data warehouse

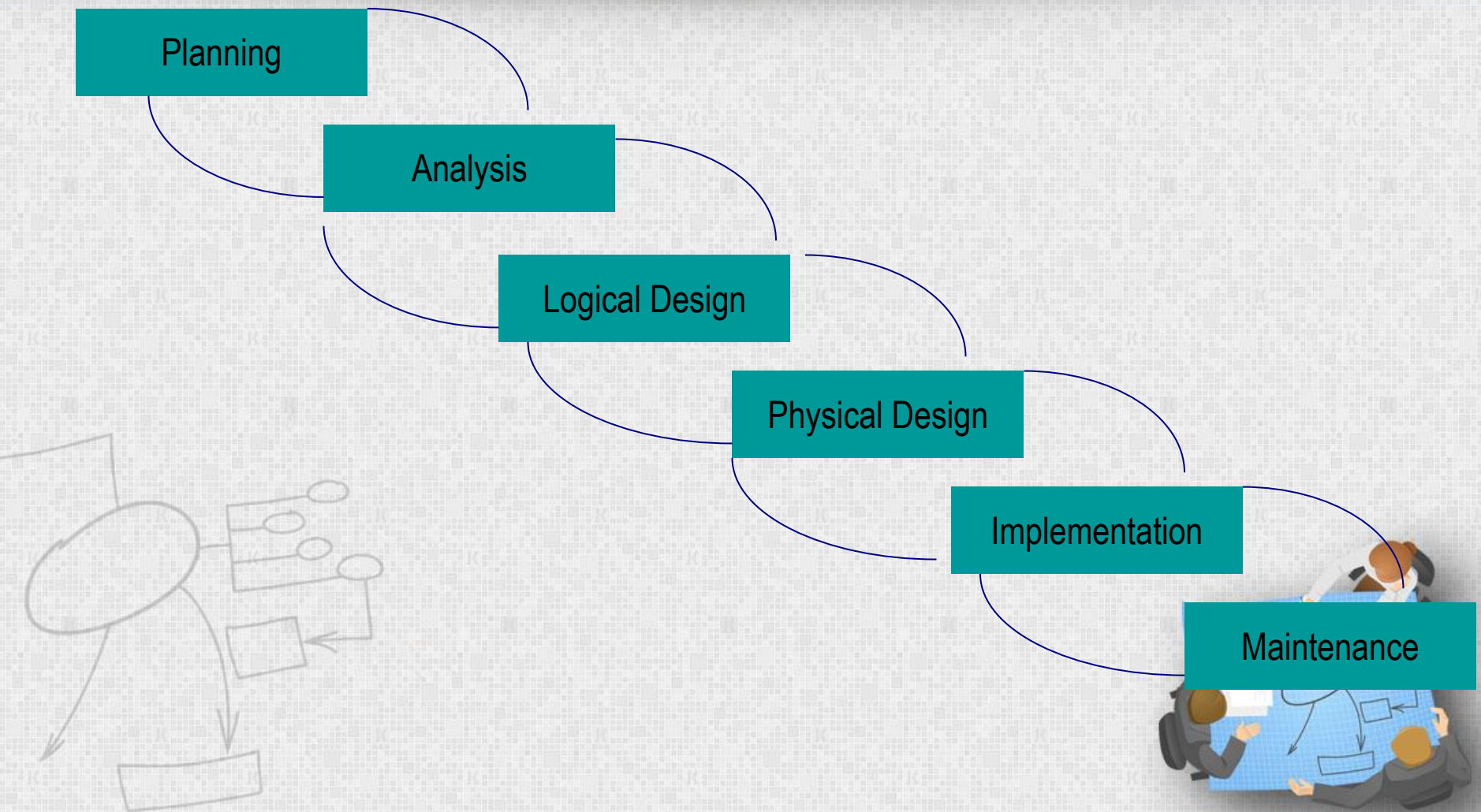


Two Approaches to Database and IS Development

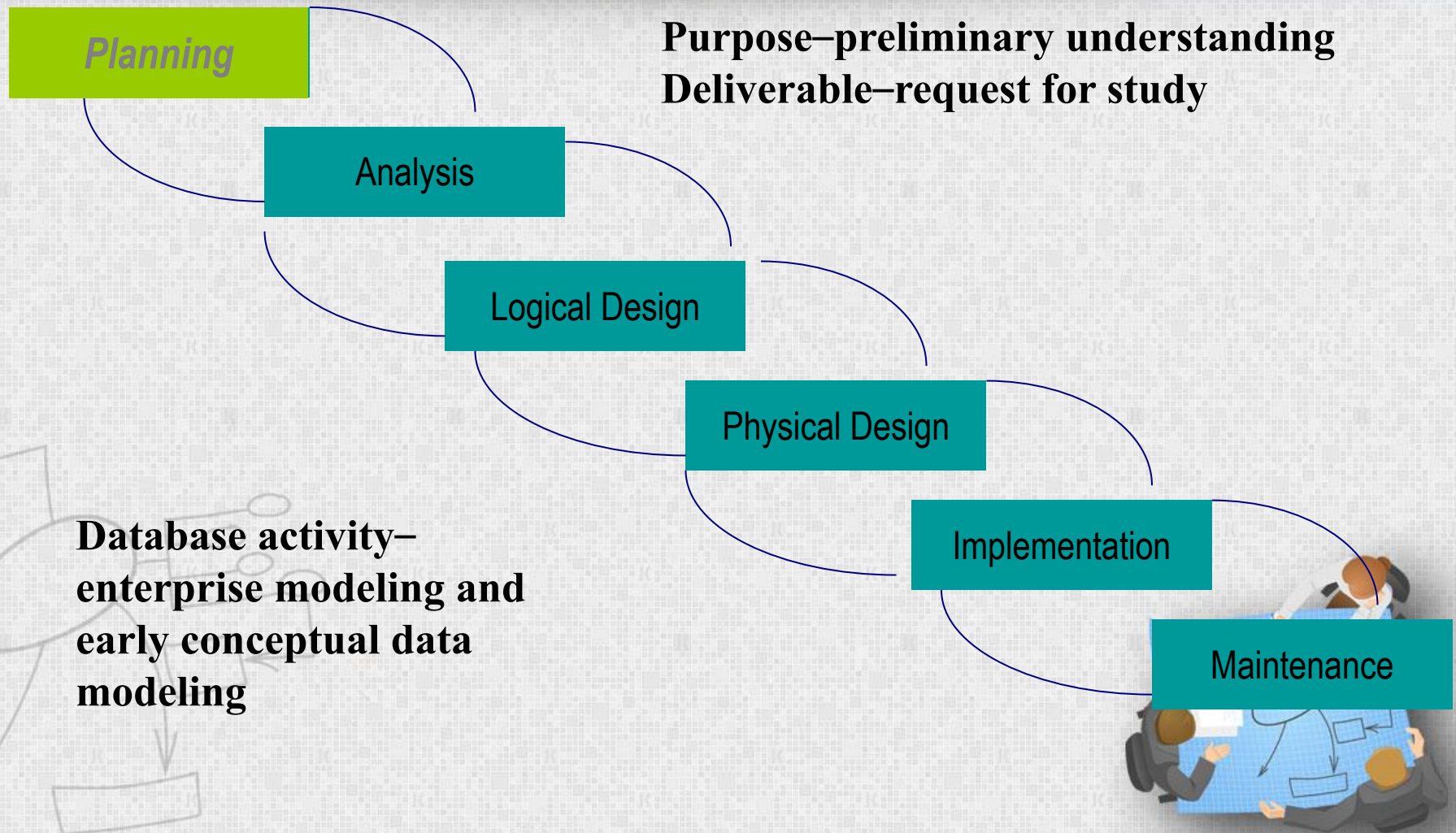
- SDLC
 - System Development Life Cycle
 - Detailed, well-planned development process
 - Time-consuming, but comprehensive
 - Long development cycle
- Prototyping
 - Rapid application development (RAD)
 - Define database during development of initial prototype
 - Repeat implementation and maintenance activities with new prototype versions



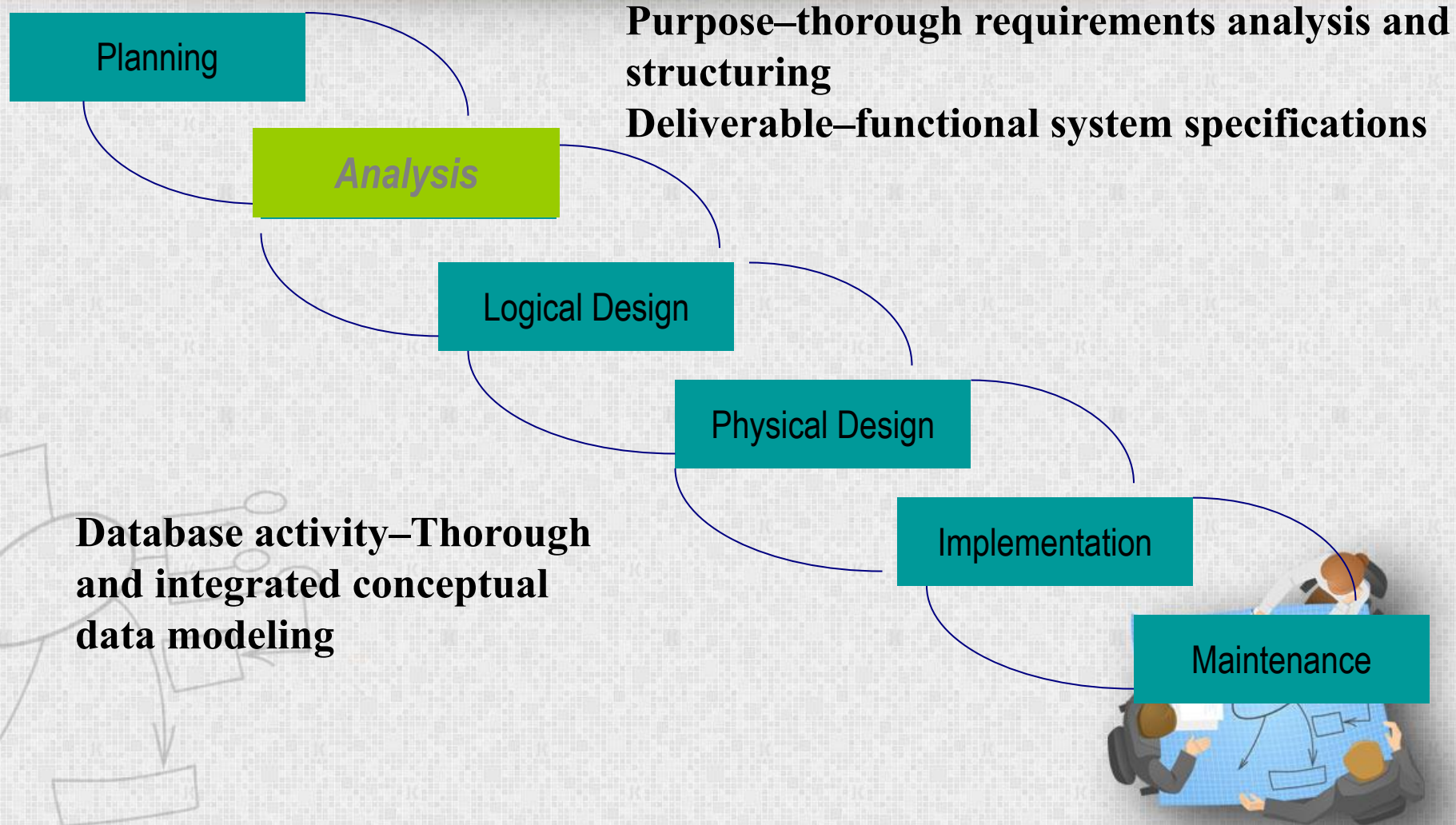
Systems Development Life Cycle



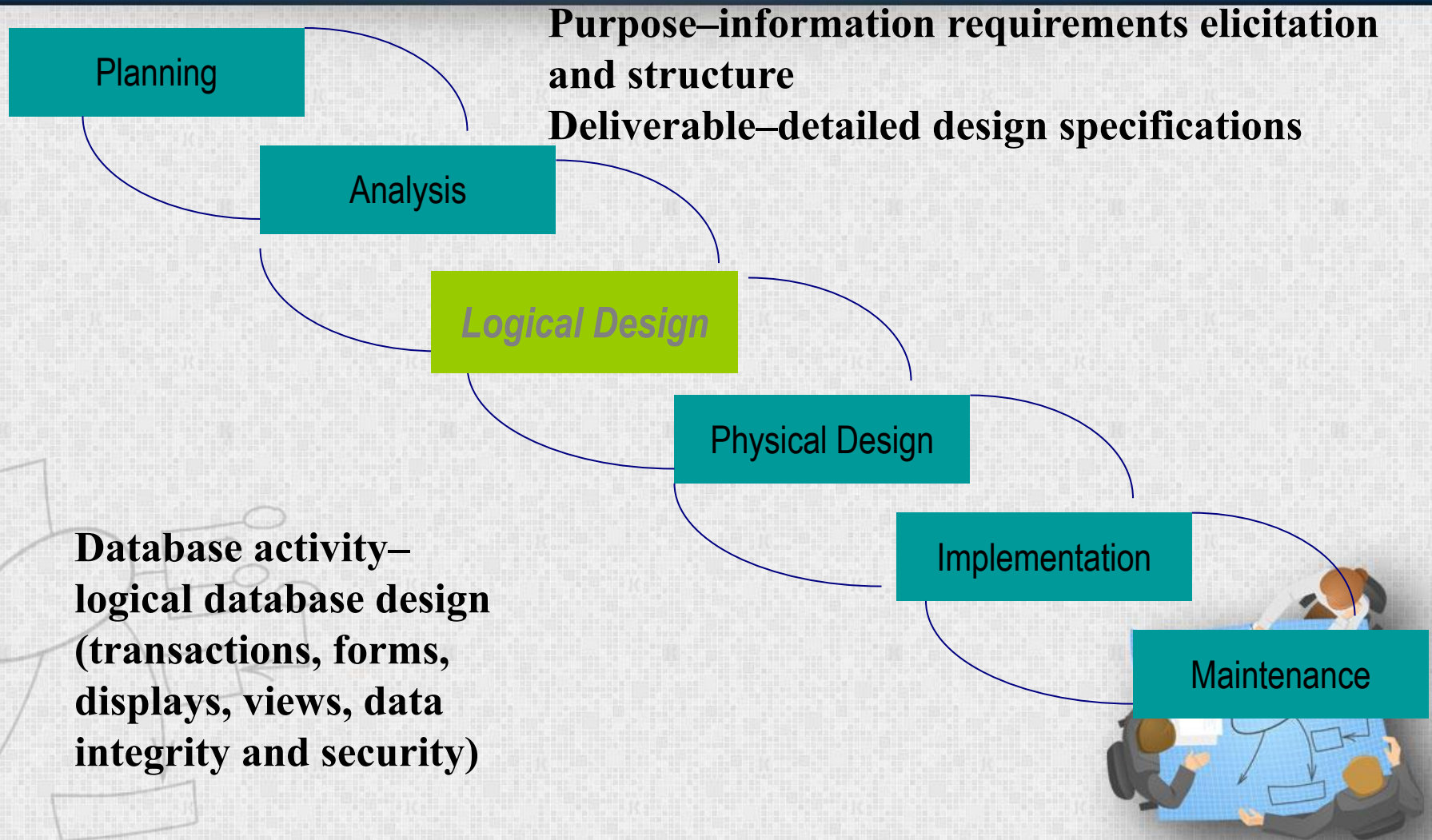
Systems Development Life Cycle



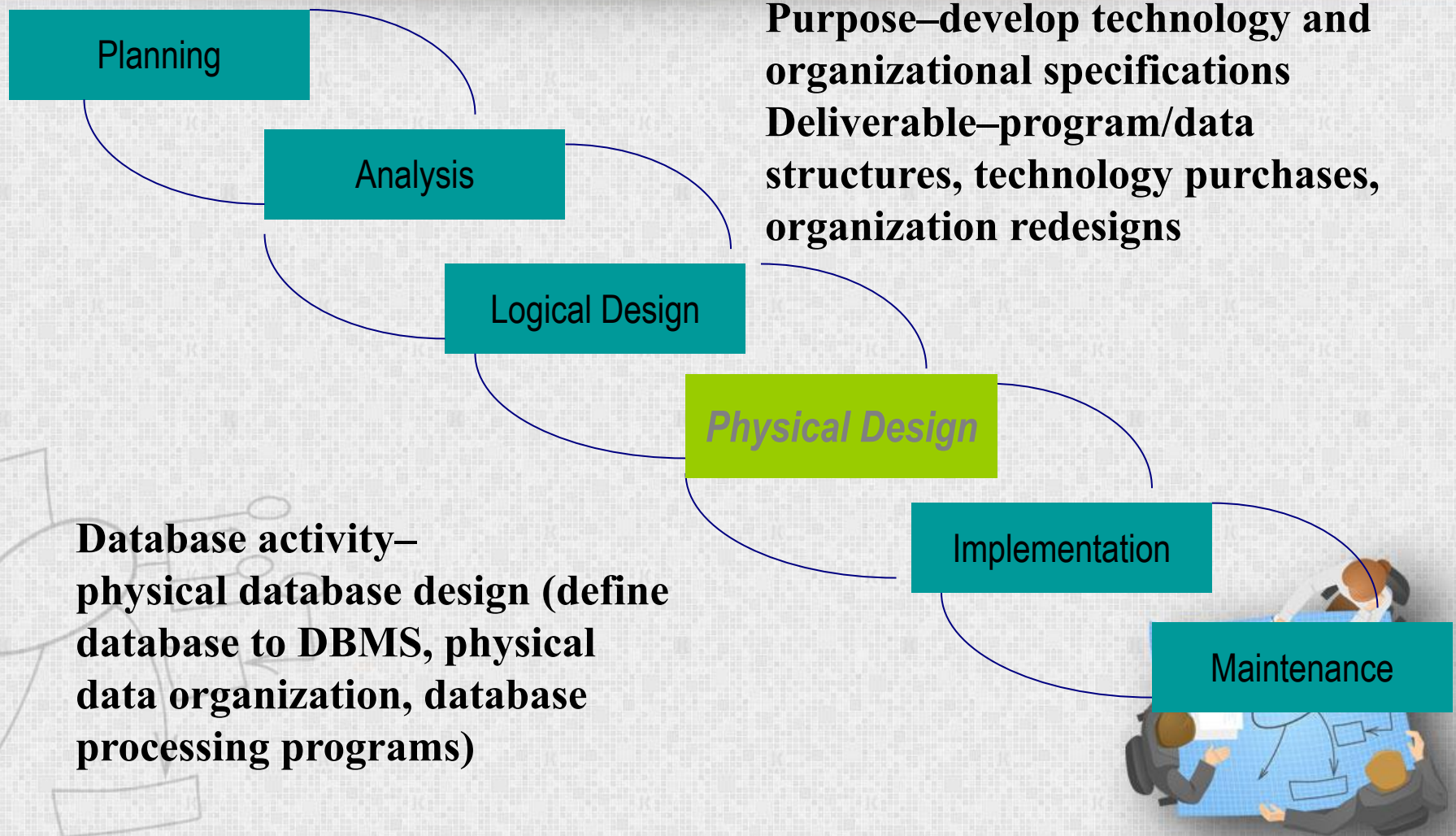
Systems Development Life Cycle (see also Figures 2.4, 2.5) (cont.)



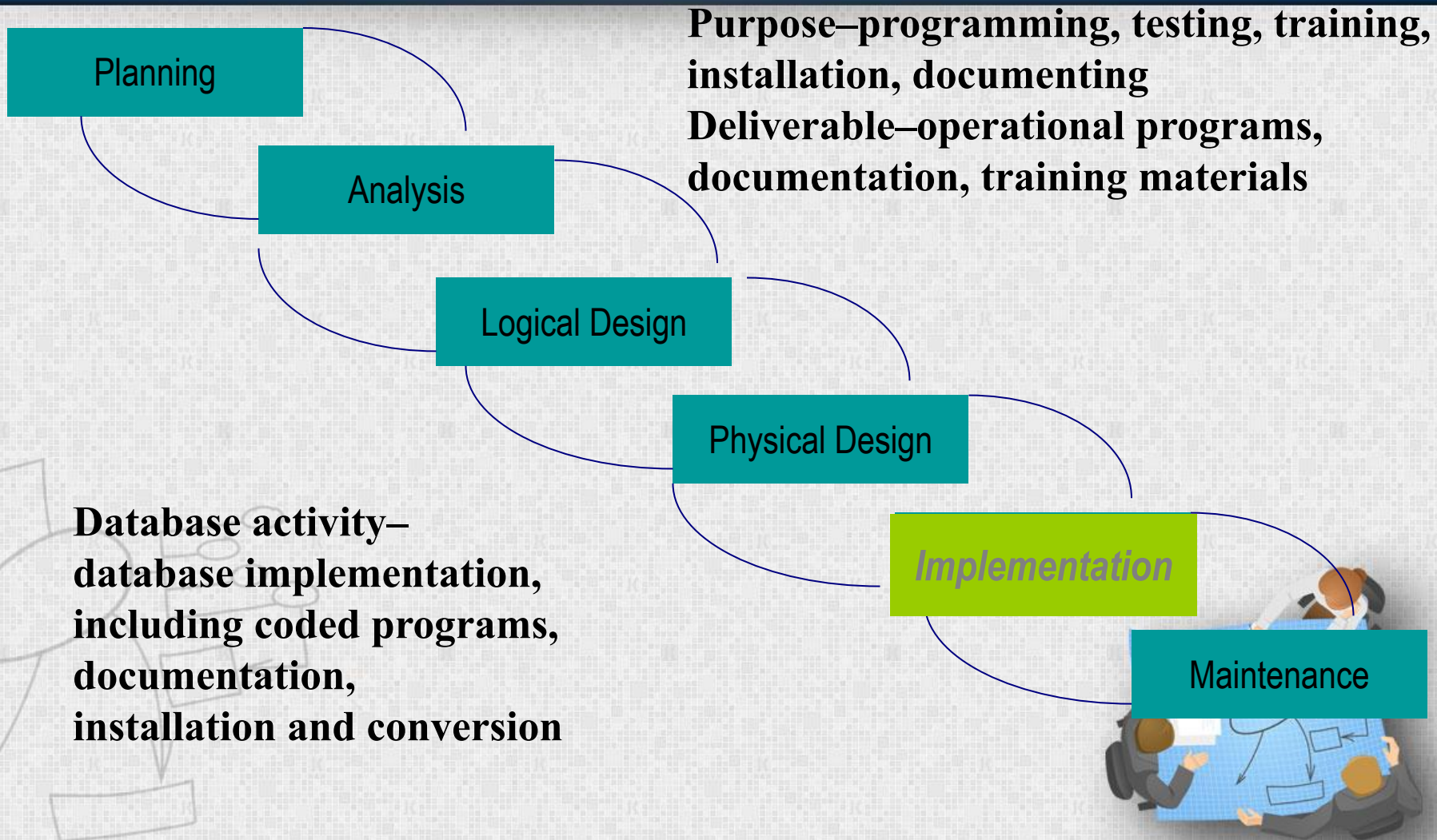
Systems Development Life Cycle



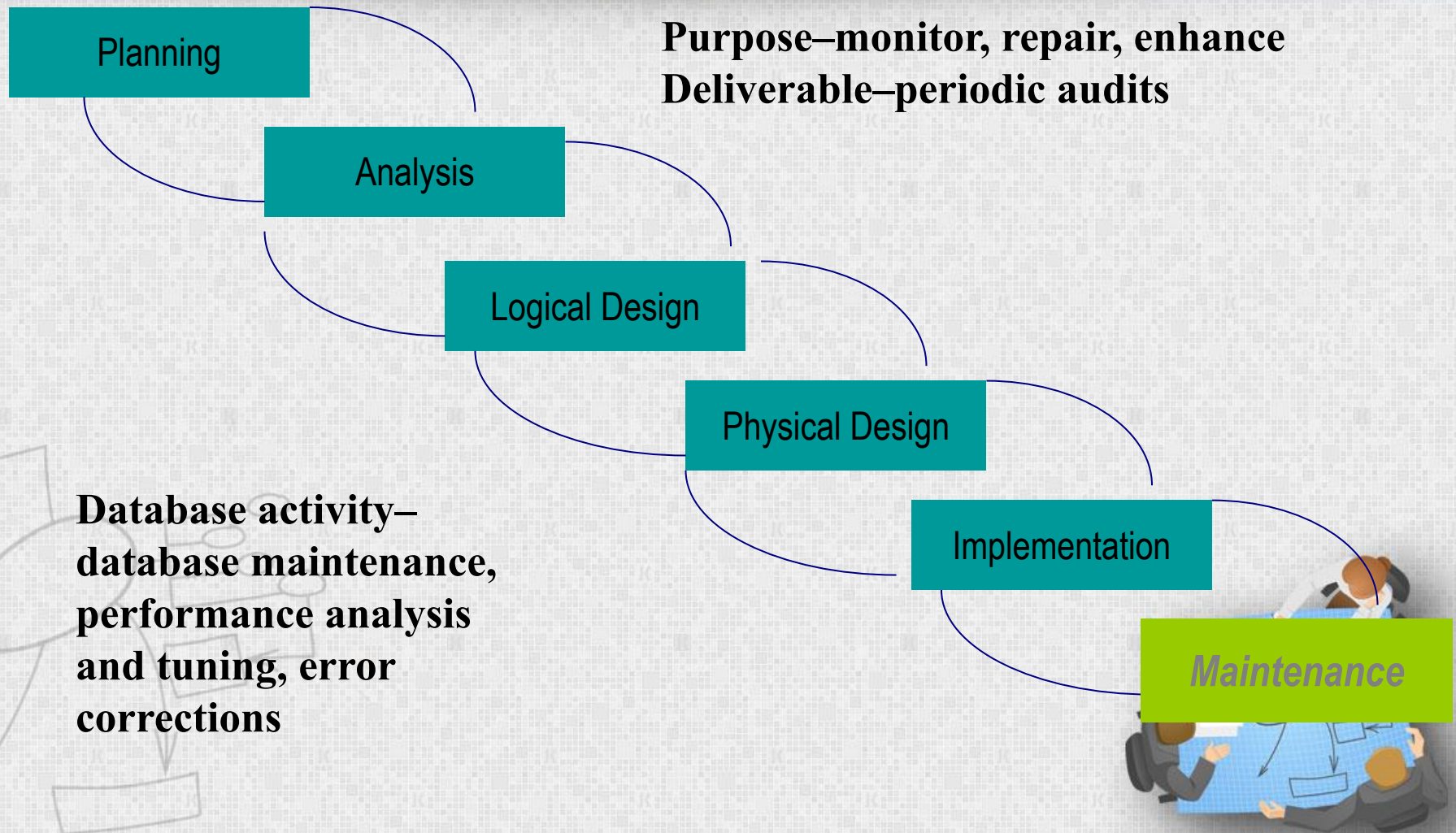
Systems Development Life Cycle



Systems Development Life Cycle



Systems Development Life Cycle



Prototyping Database Methodology

Prototyping methodology and database development process

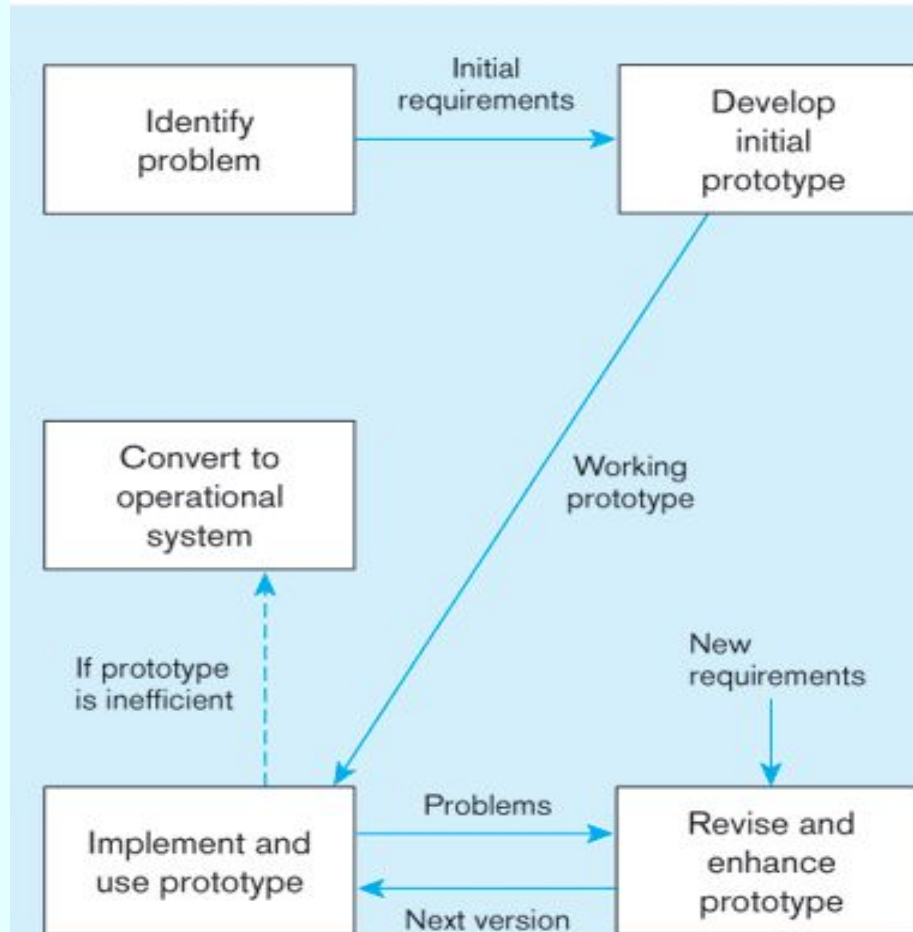


Figure 2-6 The prototyping methodology and database development process

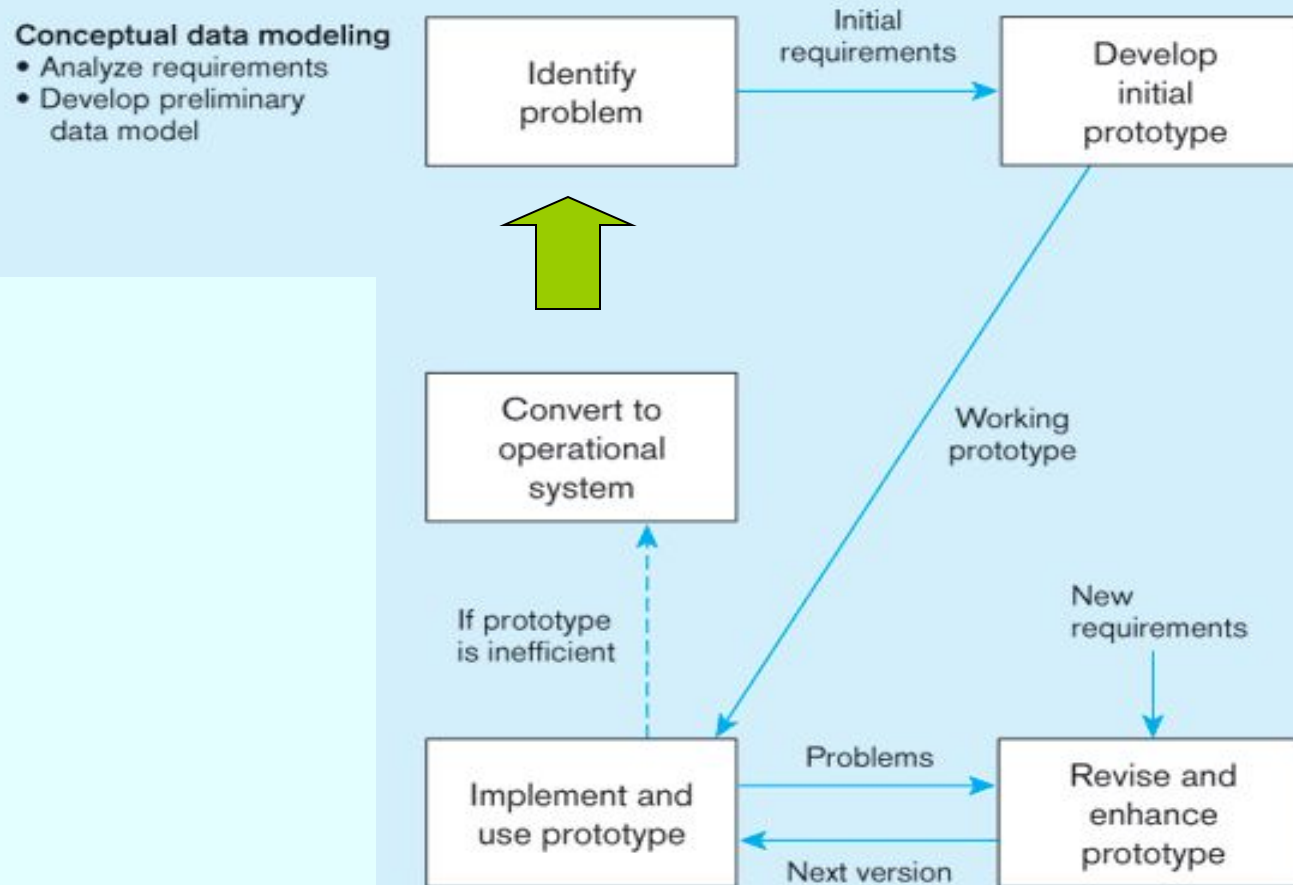


Figure 2-6 The prototyping methodology and database development process

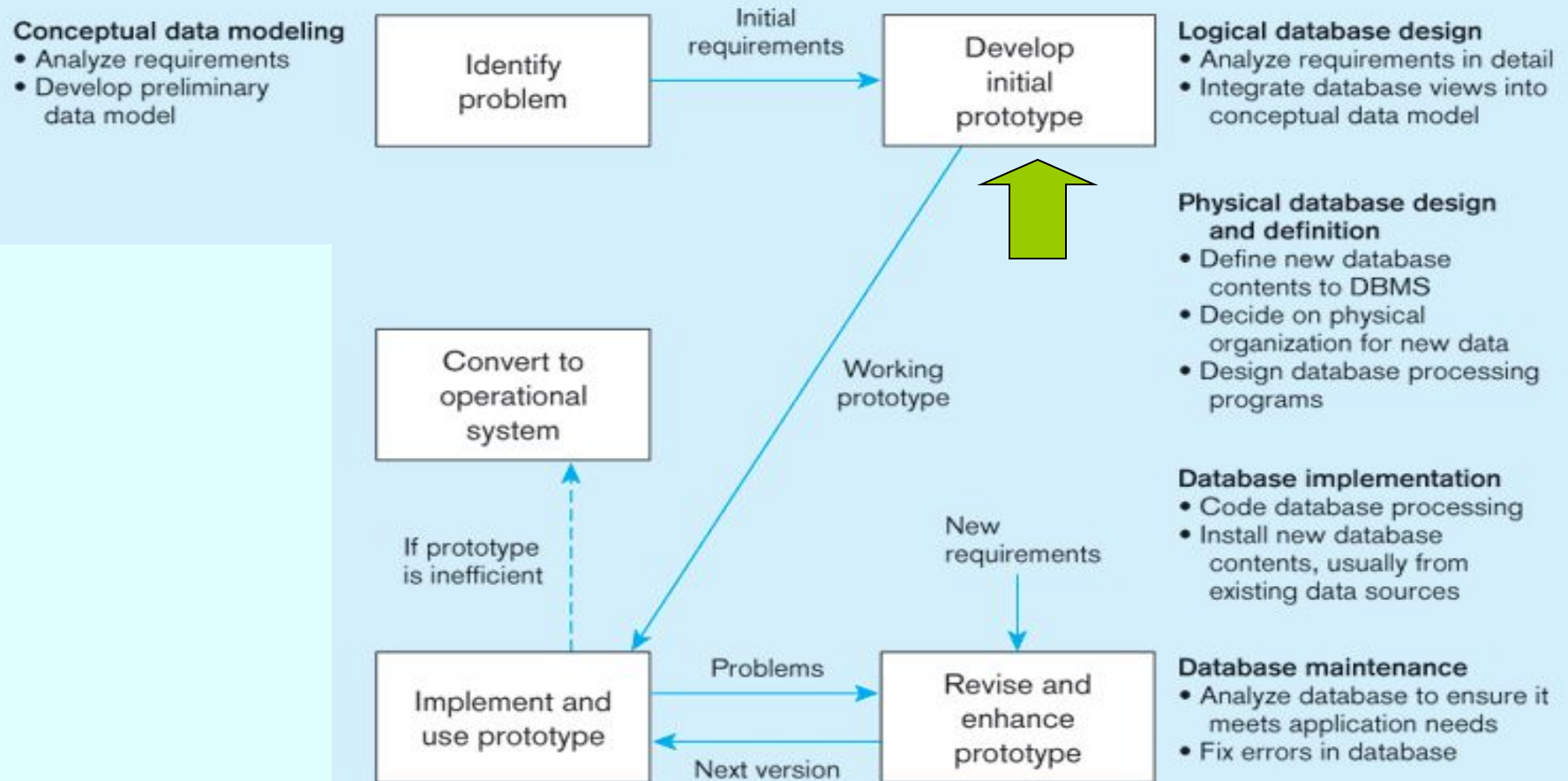


Figure 2-6 The prototyping methodology and database development process

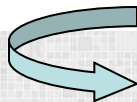
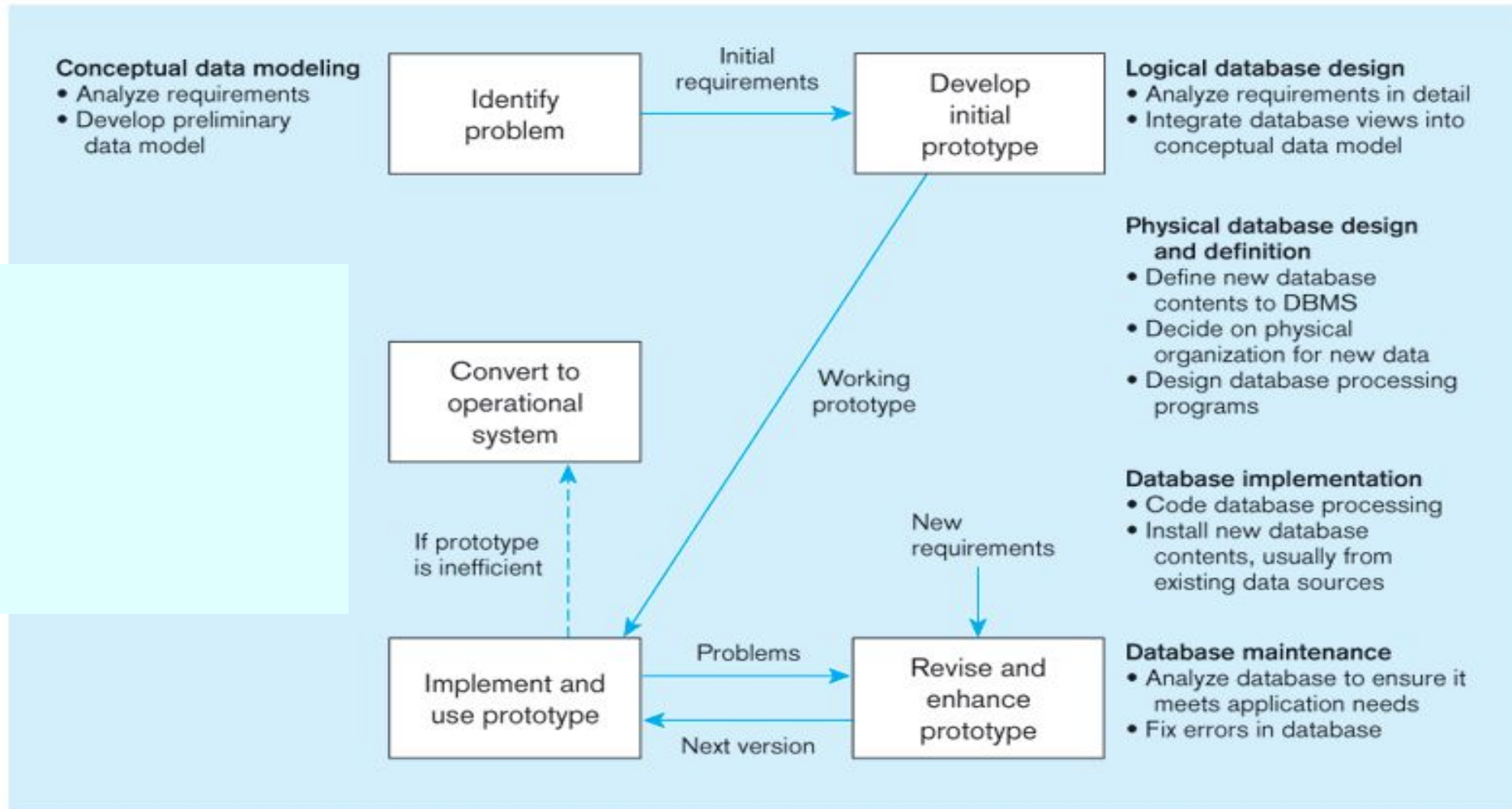
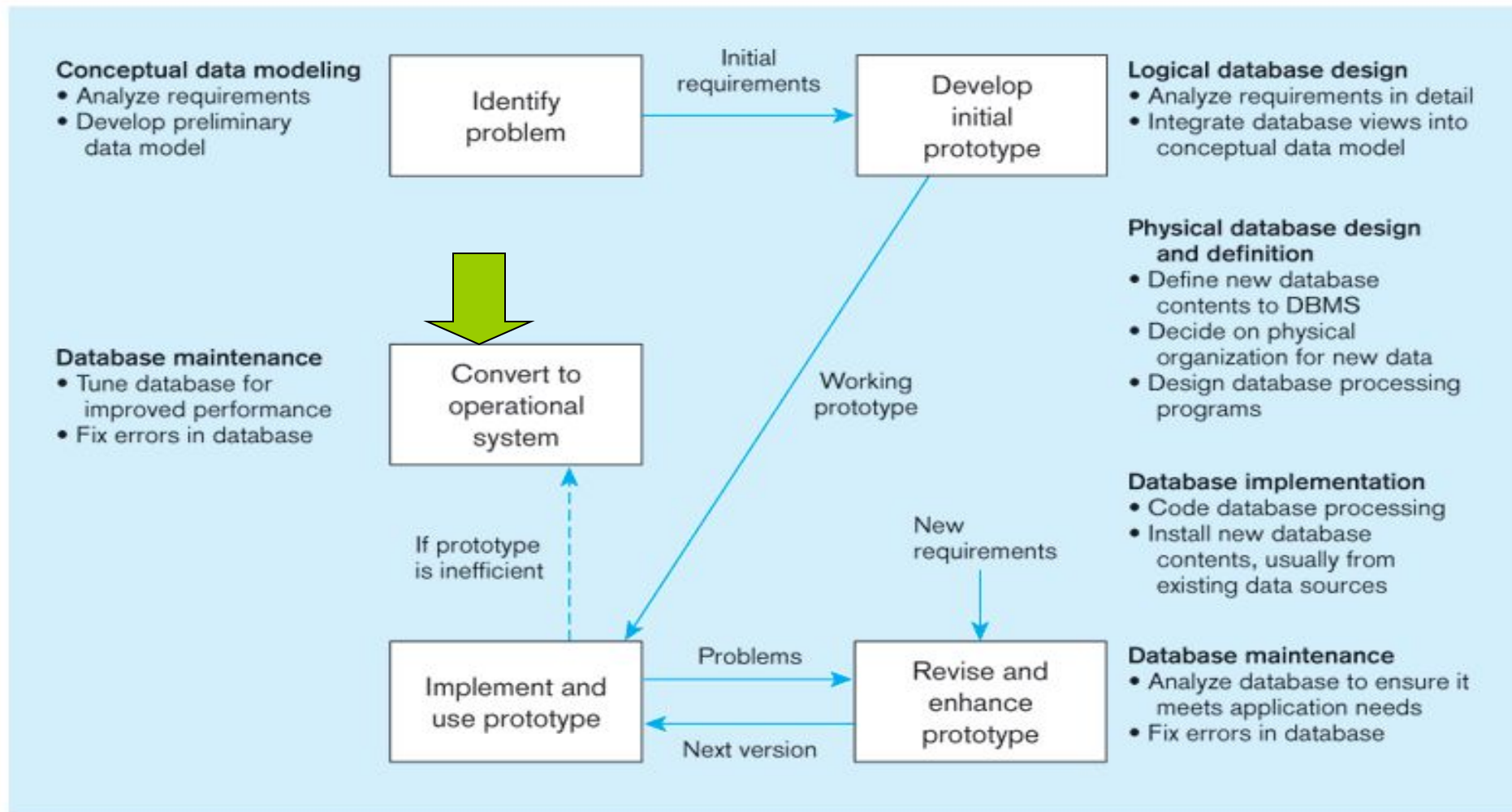


Figure 2-6 The prototyping methodology and database development process



Schema

- The word **schema** comes from the Greek word "σχῆμα" (skhēma), which means *shape*, or more generally, *plan*. The plural is "σχήματα" (skhēmata). In English, both *schemas* and *schemata* are used as plural forms, although the latter is the standard form for written English.
- **Schema** may refer to:
- Model or Diagram
 - Schematic, a diagram that represents the elements of a system using abstract, graphic symbols

Reference: <http://en.wikipedia.org/wiki/Schema>



Database Schema

- The **schema** (pronounced **skee-ma**) of a database system is its structure described in a formal language supported by the database management system (DBMS). In a relational database, the schema defines the tables, the fields, relationships, views, indexes, packages, procedures, functions, queues, triggers, types, sequences, materialized views, synonyms, database links, directories, Java, XML schemas, and other elements.
- Schemas are generally stored in a data dictionary. Although a schema is defined in text database language, the term is often used to refer to a graphical depiction of the database structure. In other words, schema is the structure of the database that defines the objects in the database.
- Levels of database schema
- Conceptual schema, a map of concepts and their relationships.
- Logical schema, a map of entities and their attributes and relations
- Physical schema, a particular implementation of a logical schema

Reference: http://en.wikipedia.org/wiki/Database_schema



Thank You

